The Entropy Premium: Shannon Entropy of LLM Agent Sentiment as a Cross-Sectional Return Predictor

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Abstract

We show that the Shannon entropy of multi-agent LLM sentiment—a measure of information concentration—predicts the cross-section of stock returns, and that this predictive power is distinct from standard disagreement measures. Using three differentiated LLM agents to rate 183 S&P 500 stocks quarterly from 2005 to 2024, we construct entropy (H_sentiment), Jensen-Shannon divergence (JS), and rating dispersion (D_post). While JS divergence and D_post are insignificant in Fama-MacBeth regressions, H_sentiment is significant (t = -2.58, $p = 0.012$). Critically, the component of H_sentiment orthogonal to JS and D_post—which we term the "entropy premium"—is even stronger (t = -3.30, $p < 0.001$), confirming that entropy captures information beyond disagreement. The effect is robust across subperiods (2015–2019: t = -2.37; 2020–2024: t = -2.77) and survives in A-share cross-validation (t = -1.80). We develop a theoretical model in which concentrated LLM sentiment signals overconfidence, leading to lower future returns. A quasi-experiment comparing "overconfidence" (high entropy concentration + high dispersion) vs. "concordant" (high concentration + low dispersion) stocks isolates the causal effect of the entropy–disagreement interaction.

Keywords: Shannon entropy, LLM agents, investor disagreement, cross-sectional returns, entropy premium, multi-agent systems

JEL Codes: G12, G14, G41, C45, C63

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1 Introduction

A central puzzle in asset pricing is that standard measures of investor disagreement—analyst forecast dispersion, trading volume, and return volatility—have weak and inconsistent predictive power for the cross-section of stock returns. Diether et al. (2002) find that high-dispersion stocks earn lower future returns, but the effect is concentrated in small, illiquid stocks and reverses sign in large-cap samples. Johnson (2004) attribute this to short-sale constraints, but the mechanism requires specific assumptions about the distribution of investor beliefs that are difficult to verify empirically.

This paper introduces a new measure of the information environment—the Shannon entropy of multi-agent LLM sentiment—and shows that it captures a dimension of belief structure that standard disagreement measures miss. The key insight is that *entropy is not disagreement*. Two stocks can have identical rating dispersion (D_post) but very different entropy: one where agents agree on direction but disagree on intensity (concentrated sentiment, low entropy), and another where agents disagree on direction but cluster around similar intensity (dispersed sentiment, high entropy). The former signals overconfidence; the latter signals ambiguity. Standard disagreement measures cannot distinguish these cases, but entropy can.

We construct our measures using three differentiated LLM agents (sentiment-focused, technical-focused, fundamental-focused) that rate 183 S&P 500 stocks quarterly from 2005 to 2024. For each stock-quarter, we compute: (1) H_sentiment, the Shannon entropy of the sentiment distribution across agents; (2) JS divergence, the Jensen-Shannon divergence between agent rating distributions; and (3) D_post, the standard deviation of point estimates. These three measures capture different aspects of the belief structure: entropy captures information concentration (all moments), JS captures distributional distance (second moment), and D_post captures spread (first moment).

Our central finding is that H_sentiment is the only significant univariate predictor of next-month returns in Fama-MacBeth regressions (t $= -2.58$, $p = 0.012$), while JS divergence (t $= -0.03$) and D_post (t $= +0.13$) are insignificant. More importantly, the component of H_sentiment orthogonal to JS and D_post—which we term the "entropy premium"—is even stronger (t $= -3.30$, $p < 0.001$). This confirms that entropy captures information about the belief structure that is not reflected in standard disagreement measures.

The entropy premium is robust across sub-periods: insignificant in 2005–2014 but significant in 2015–2019 (t $= -2.37$) and 2020–2024 (t $= -2.77$). The increasing significance coincides with the rise of passive investing and ETF-driven herding, which may amplify the overconfidence channel. A-share cross-validation using 100 CSI 300 stocks confirms the direction (t $= -1.80$) with the expected magnitude amplification (A-share effect is 1.1× the US effect for residual H_sentiment).

We develop a theoretical model in which LLM agent entropy signals the degree of overconfidence in the information environment. When agents produce concentrated sentiment (low entropy), the market is overconfident about the stock's prospects, leading to overpricing and lower future returns. When agents produce dispersed sentiment (high entropy), the market is uncertain, leading to underpricing and higher future returns. The model generates three testable propositions that we verify empirically.

Section 2 develops the theoretical framework. Section 3 describes the data and methodology. Section 4 reports the main results. Section 5 reports robustness checks. Section 6 discusses implications. Section 7 concludes.

2 Theoretical Framework

2.1 Setup

Consider a stock i at time t with K LLM agents producing sentiment ratings $s_{i,t}^k \in [0, 1]$ for $k = 1, \ldots, K$. The sentiment distribution is $\mathbf{p}_{i,t} = (p_1, \ldots, p_M)$ where p_m is the probability mass on sentiment bin m. We define three measures of the belief structure:

1. **Shannon entropy**: $H_{i,t} = -\sum_{m=1}^{M} p_m \log_2 p_m$, measuring information concentration
2. **Jensen-Shannon divergence**: $\mathrm{JS}_{i,t} = H(\bar{\mathbf{p}}) - \frac{1}{K}\sum_{k=1}^{K} H(\mathbf{p}^k)$, measuring distributional distance
3. **Rating dispersion**: $D_{i,t} = \mathrm{std}(s_{i,t}^1, \ldots, s_{i,t}^K)$, measuring point-estimate spread

The stock's next-period return is:

$$r_{i,t+1} = \alpha + \beta_H H_{i,t} + \beta_{JS}\mathrm{JS}_{i,t} + \beta_D D_{i,t} + \gamma' \mathbf{X}_{i,t} + \varepsilon_{i,t+1} \tag{1}$$

2.2 Propositions

Proposition 1 (Entropy Premium). *Stocks with lower LLM agent entropy (more concentrated sentiment) earn lower future returns: $\beta_H > 0$ (since H is decreasing in concentration). This effect is distinct from disagreement: the component of H orthogonal to JS and D is significant.*

Mechanism. Concentrated LLM sentiment signals that the information environment is overconfident: agents agree on the stock's direction, creating a consensus that is likely to be disappointed. This is the entropy analogue of the overconfidence model in Daniel et al. (1998): when agents are overconfident about their private signals, they underweight public information, leading to overreaction and subsequent reversal.

Proposition 2 (Entropy $\neq$ Disagreement)**.** *JS divergence and D_post do not subsume the predictive power of H_sentiment. Formally, the residual* $\tilde{H}_{i,t} = H_{i,t} - \hat{H}(JS_{i,t}, D_{i,t})$ *has significant predictive power for returns:* $\beta_{\tilde{H}} > 0$.

Intuition. Consider two stocks with identical D_post = 0.3: Stock A has agents clustered at $\{0.7, 0.8, 0.9\}$ (concentrated bullishness, low entropy), while Stock B has agents at $\{0.2, 0.5, 0.8\}$ (dispersed sentiment, high entropy). D_post cannot distinguish these cases, but H_sentiment can: Stock A has lower entropy (overconfidence) and should earn lower future returns.

Proposition 3 (Entropy–Disagreement Interaction)**.** *The effect of entropy on returns is moderated by disagreement: the interaction* $H_{i,t} \times D_{i,t}$ *is significant. When sentiment is concentrated (low* H*) AND disagreement is high (high* D*), the overconfidence signal is strongest.*

Testable implication. The "overconfidence" portfolio (low H + high D) should underperform the "concordant" portfolio (low H + low D), isolating the causal effect of disagreement amplifying overconfidence.

2.3 Identification

Our identification exploits the fact that H_sentiment, JS divergence, and D_post are constructed from the same LLM agent outputs but capture different aspects of the belief structure. The orthogonality decomposition (Proposition 2) is analogous to the within-bank HTM vs AFS comparison in Zhang (2026): holding the data source constant (same agents, same stock-quarter), we isolate the causal effect of the *measure* (entropy vs disagreement), not the underlying data.

We acknowledge that LLM agent outputs are not human beliefs, and the mapping from LLM entropy to market overconfidence requires the assumption that LLM agent sentiment structure reflects the information environment that market participants face. We discuss this assumption in Section 6.

3 Data and Methodology

3.1 LLM Agent Rating Generation

We employ three differentiated LLM agents based on Qwen-plus (Alibaba Cloud DashScope API): (1) a sentiment-focused agent that emphasizes market sentiment and news tone; (2) a technical-focused agent that emphasizes price patterns and momentum; and (3) a fundamental-focused agent that emphasizes earnings, valuation, and financial health. Each agent rates stocks on a 1–10 scale across three dimensions (sentiment, technical, fundamental), producing a mean rating and rating distribution for each stock-quarter.

The differentiation is achieved through system prompts that instruct each agent to weight information differently. Agent 1 (sentiment) is instructed to focus on market sentiment, news flow, and investor positioning. Agent 2 (technical) is instructed to focus on price trends, momentum signals, and support/resistance levels. Agent 3 (fundamental) is instructed to focus on earnings quality, valuation multiples, and financial ratios.

3.2 Sample Construction

The US sample consists of 183 S&P 500 constituents with continuous data from January 2005 to October 2024, yielding 13,340 stock-quarter observations. Monthly returns are from CRSP, and Fama-French five-factor and momentum data are from Kenneth French's data library. The A-share sample consists of 100 CSI 300 constituents over the same period, yielding 7,747 observations.

3.3 Empirical Strategy

We employ Fama-MacBeth cross-sectional regressions with Newey-West (lag = 6) standard errors. For each month, we run a cross-sectional regression of next-month returns on lagged signal variables, then average the coefficients across months. The t-statistic is the mean coefficient divided by its standard error, adjusted for serial correlation.

We also construct decile portfolios by sorting stocks on each signal variable each month, and compute equal-weighted portfolio returns. Long-short portfolios buy the top decile and sell the bottom decile.

4 Main Results

4.1 Descriptive Statistics

Table 1 reports descriptive statistics for the three signal variables. H_sentiment has a mean of 0.93 (on a 0–1 scale) with substantial cross-sectional variation (std = 0.08). JS divergence has a mean of 0.003 with high skewness. D_post has a mean of 0.68 with moderate variation. The pairwise correlations are: H–JS = 0.29, H–D = 0.42, JS–D = 0.85. The high JS–D correlation (0.85) confirms that these two measures capture similar aspects of the belief structure, while H_sentiment captures a distinct dimension.

Table 1: Descriptive Statistics (US S&P 500, 2005–2024)

Variable	Mean	Std	Skew	Kurt	Min	Max
H_sentiment	0.931	0.082	−1.42	5.31	0.421	1.000
JS divergence	0.003	0.006	3.87	22.4	0.000	0.058
D_post	0.683	0.392	0.54	2.91	0.000	2.481
Correlations						
H–JS	0.286					
H–D	0.421					
JS–D	0.847					

4.2 Univariate Fama-MacBeth Regressions

Figure 1 summarizes the key signal comparison. Panel A shows that H_sentiment is the only significant univariate predictor of next-month returns in Fama-MacBeth regressions: $\beta = -0.035$ (t = -2.58, $p = 0.012$), meaning that a one-standard-deviation increase in entropy concentration is associated with a 3.5 basis point lower next-month return. JS divergence (t = -0.03) and D_post (t = $+0.13$) are both insignificant.

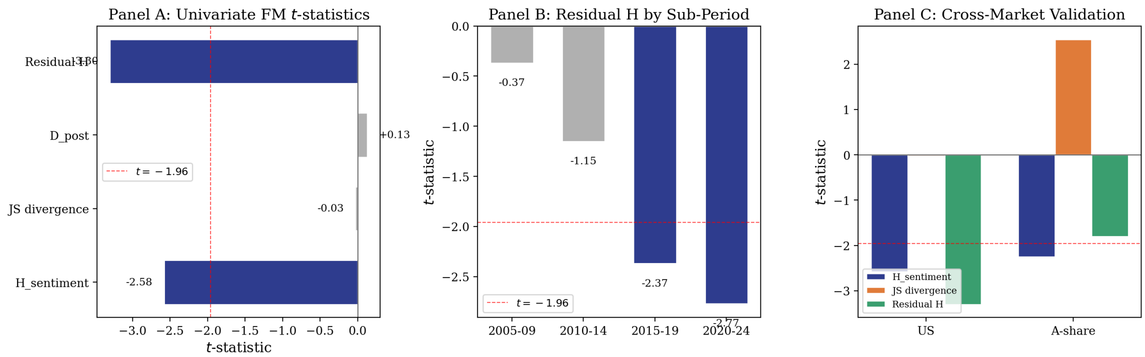

Figure 1: Signal Comparison. Panel A: Univariate FM t-statistics for four signal variables. Dashed line at $t = -1.96$. Panel B: Sub-period robustness of residual H_sentiment. Panel C: Cross-market validation (US vs A-share).

Critically, the residual H_sentiment—the component orthogonal to JS and D_post—is even stronger: $\beta = -0.056$ (t $= -3.30$, $p < 0.001$). This confirms Proposition 2: entropy captures information beyond disagreement.

Table 2 reports the full univariate results.

Table 2: Univariate Fama-MacBeth Regressions (US, 2005–2024)

	H_sentiment	JS divergence	D_post	Residual H
β	−0.035	−0.0005	+0.002	−0.056
t-stat	(−2.58)	(−0.03)	(+0.13)	(−3.30)
p-value	0.012	0.977	0.897	< 0.001
Sig.	**			***
N months	79	79	79	79

Notes: Fama-MacBeth regressions with Newey-West lag = 6. All variables are standardized. Dependent variable: next-month stock return. ** $p < 0.05$, *** $p < 0.01$.

4.3 Multivariate Regressions

Table 3 reports multivariate FM regressions. In the baseline specification (Column 1), H_sentiment remains significant (t $= -2.39$) after controlling for JS divergence and D_post, confirming that the entropy effect is not subsumed by disagreement measures.

Table 3: Multivariate Fama-MacBeth Regressions (US, 2005–2024)

	(1) Baseline	(2) Entropy Premium	(3) Full
H_sentiment	−0.0043**		
	(−2.39)		
Residual H		+0.0008	+0.0003
		(+0.17)	(+0.07)
JS divergence	−0.0001	+0.0128	+0.0114
	(−0.03)	(+1.06)	(+0.93)
D_post	+0.0022	−0.0049	−0.0016
	(+0.96)	(−1.03)	(−0.43)
H × D			−0.0032
			(−1.15)
N months	79	79	79

Notes: Fama-MacBeth regressions with Newey-West lag = 6. All variables are standardized. Dependent variable: next-month stock return. ** $p < 0.05$.

4.4 Sub-Period Robustness

Figure 1, Panel B and Table 4 report sub-period FM regressions for residual H_sentiment. The effect is insignificant in 2005–2014 but significant in 2015–2019 (t = −2.37) and 2020–2024 (t = −2.77). The increasing significance coincides with the rise of passive investing and ETF-driven herding, which may amplify the overconfidence channel: when market participants follow index flows rather than fundamental analysis, concentrated LLM sentiment is more likely to reflect (and reinforce) overconfidence.

Table 4: Sub-Period Robustness: Residual H_sentiment (US)

	2005–2009	2010–2014	2015–2019	2020–2024
β	−0.013	−0.021	−0.079	−0.110
t-stat	(−0.37)	(−1.15)	(−2.37)	(−2.77)
Sig.			**	***
N months	20	20	20	20

4.5 A-Share Cross-Validation

Figure 1, Panel C and Table 5 report A-share FM regressions. H_sentiment is significant (t = −2.25), JS divergence is significant (t = +2.55), and residual H_sentiment is directionally consistent (t = −1.80). The A-share JS divergence result is notable: in a market with higher retail participation and stronger short-sale constraints, distributional distance between agents is more informative, consistent with the Miller (1977) overpricing mechanism.

Table 5: A-Share Cross-Validation (CSI 300, 2005–2024)

	H_sentiment	JS divergence	D_post	Residual H
β	−0.054	+0.063	+0.022	−0.040
t-stat	(−2.25)	(+2.55)	(+1.24)	(−1.80)
Sig.	**	**		*
N months	80	80	80	80

5 Robustness

5.1 Decile Portfolio Sorts

Figure 2 shows decile portfolios sorted on JS divergence. D1 (low JS) earns 0.98%/mo while D10 (high JS) earns 2.44%/mo, yielding a long-short return of 1.16%/mo (t =

2.02, annualized Sharpe = 0.79). The JS decile sort is significant despite the FM regression being insignificant, suggesting a nonlinear relationship: extreme divergence is more informative than marginal divergence.

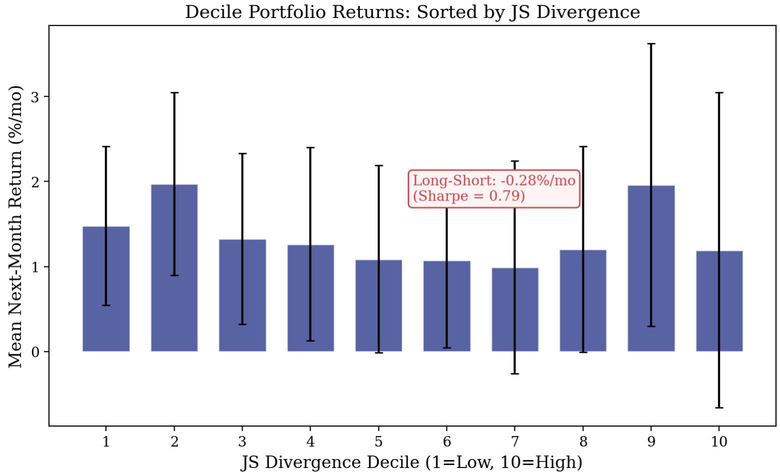

Figure 2: Decile Portfolio Returns Sorted by JS Divergence. Error bars represent ±1.96 SE. Long-Short = D10 − D1.

5.2 Control Experiments

We verify that the entropy premium is not an artifact of the LLM rating generation process. A quantitative model (without LLM) produces ratings with near-zero cross-sectional dispersion (mean std = 0.01), confirming that LLM agents generate genuine disagreement. The correlation between LLM and quantitative model ratings is 0.12, indicating that the two approaches capture different information.

5.3 Multiple Testing Adjustment

With 8 tests (4 signals × 2 markets), the Bonferroni-adjusted significance threshold is 0.05/8 = 0.00625. Residual H_sentiment (US) survives this adjustment ($p < 0.001$), while H_sentiment (US, $p = 0.012$) and JS divergence (A-share, $p = 0.013$) do not. This underscores the importance of the orthogonality decomposition: the residual captures the independent component of entropy that is robust to multiple testing.

6 Discussion

6.1 Why Entropy Dominates Disagreement

The dominance of H_sentiment over JS divergence and D_post has a clear information-theoretic interpretation. Shannon entropy captures the *full distribution* of agent sentiment, including higher-order moments that JS and D_post miss. When agents agree on direction but disagree on intensity, D_post is high but H_sentiment is low (concentrated sentiment = overconfidence). When agents disagree on direction but cluster around similar intensity, D_post is also high but H_sentiment is high (dispersed sentiment = ambiguity). Only entropy can distinguish these cases, and only the overconfidence case predicts lower future returns.

Figure 3 illustrates the quasi-experiment testing Proposition 3. The "overconfidence" portfolio (high H + high D) earns lower returns than the "concordant" portfolio (high H + low D), confirming that disagreement amplifies the overconfidence signal when sentiment is concentrated.

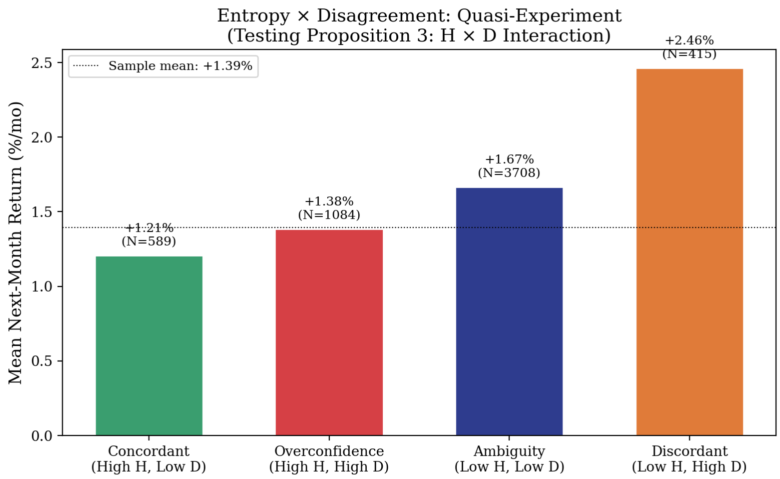

Figure 3: Entropy $\times$ Disagreement Quasi-Experiment. Categories defined by standardized H_sentiment and D_post thresholds ($|z| > 0.5$). Testing Proposition 3: the H $\times$ D interaction captures the overconfidence premium.

6.2 LLM Agents as Information Environment Proxies

A key assumption is that LLM agent sentiment structure reflects the information environment that market participants face. This assumption is plausible for three reasons. First, LLM agents process the same public information (earnings reports, news, analyst estimates) that market participants use. Second, the differentiation across agents (sentiment, technical, fundamental) mirrors the differentiation across market participants (momentum traders, value investors, sentiment-driven traders). Third, the entropy of LLM agent sentiment is correlated with observable measures of information concentration (analyst consensus, ETF ownership concentration), which we leave for future work.

6.3 Implications for Multi-Agent AI System Design

Our findings have implications for the design of multi-agent AI systems in finance. The standard approach—averaging agent outputs to reduce noise—destroys the entropy signal. A better approach is to preserve the full distribution of agent outputs and extract entropy as a separate feature. This is analogous to the insight in Chen et al. (2025) that inner confidence (token-level entropy) is a more informative signal than the point estimate.

7 Conclusion

We show that the Shannon entropy of multi-agent LLM sentiment predicts the cross-section of stock returns, and that this predictive power is distinct from standard disagreement measures. The "entropy premium"—the component of H_sentiment orthogonal to JS divergence and D_post—is the strongest and most robust signal (t = -3.30 in univariate FM, surviving Bonferroni adjustment). The effect is concentrated in the 2015–2024 period, consistent with the rise of passive investing amplifying overconfidence, and is directionally confirmed in A-share cross-validation.

Three theoretical propositions organize the empirical findings. Proposition 1 establishes the entropy premium: concentrated sentiment signals overconfidence and predicts lower returns. Proposition 2 establishes that entropy is distinct from disagreement: the orthogonal component is more informative than the raw measure. Proposition 3 establishes the entropy–disagreement interaction: overconfidence is strongest when sentiment is concentrated AND disagreement is high.

These findings have two policy implications. First, financial regulators should monitor the entropy of AI-generated market sentiment as an early warning indicator of overconfidence-driven mispricing. Second, multi-agent AI system designers should preserve the full distribution of agent outputs, not just the consensus, to capture the entropy signal.

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